Kazakh-British Technical University

Data collection and preparation

STUDENTS' INDEPENDENT STUDY (SIS) №1

The influence of budget and critical reception on major film awards

Full Name: Sayin Bekarys,

Sailau Maksat, Tlek Margulan

ID: 23B030392, 23B031422,

23B030315.

08.10.2026

Almaty

1. Introduction and Research Objectives

The goal of this project is to determine whether there is a relationship between professional film critics' ratings (Metascore), audience ratings (IMDb), and the commercial success (box office receipts) of Oscar-winning films.

2. Methodology and Data Preparation

A complete data preparation cycle was executed for this study:

- Data Collection: Data was gathered from two independent sources. A list of 1,401 Oscar-winning films (including titles and release years) was obtained via web scraping from a Wikipedia page. This dataset was then enriched with financial metrics (BoxOffice), critic ratings (Metascore), and audience scores (IMDb) using the OMDb REST API.
- Data Cleaning: String-based financial values and ratings were stripped of extraneous characters (such as dollar signs and commas) and converted into a numeric format (float64).
- Final Dataset: After removing duplicates and records with missing key metrics (primarily older films lacking historical financial data in the API), the final clean dataset consisted of 496 records.

3. Analysis Results

3.1. Relationship Between Ratings and Box Office (Scatter Plot)

Visual analysis revealed that the majority of Oscar-winning films gross under $200 million, though a few exceptional blockbusters reach up to $790 million. There is no clear dependency between critic scores and box office revenue:

- Films with the highest critic ratings (Metascore 80–100) are evenly distributed across the revenue spectrum, encompassing both low-budget indie films and massive blockbusters.
- Conversely, films with lower critic ratings (40–60) occasionally demonstrate outstanding commercial results.

3.2. Distribution of Audience Ratings (Box Plot)

The analysis of IMDb ratings distribution highlighted a consistently positive audience reception for Oscar-winning films:

- The median audience rating is notably high at 7.6 out of 10.
- The vast majority of the films (the interquartile range) fall into a very narrow and high-scoring interval from 7.2 to 8.0.
- Ratings below 6.0 are extremely rare, with only two outlier films identified (scoring 5.7 and 5.9).

3.3. Correlation Analysis

Calculating Pearson correlation coefficients yielded the following insights:

- Metascore and IMDb ($r \approx 0.57$): *Moderate positive correlation.* The opinions of professional critics and the general audience generally align.
- Box Office and IMDb ($r \approx 0.18$): *Weak positive correlation.* High audience ratings are only slightly associated with higher box office receipts.

- Box Office and Metascore ($r \approx 0.00$): *No correlation.* The ratings given by the professional press have absolutely no linear relationship with a film's financial success.

4. Key Findings
    1. Quality Does Not Guarantee Super Profits: Within this sample, critical acclaim is not a driver of box office revenue. The commercial success of Oscar-winning films is likely dependent on other factors—such as marketing budgets, genre appeal, and distribution scale—rather than press reviews.
    2. Audience and Expert Consensus: While critics do not influence the box office, their assessment of a film's quality generally matches that of the viewers. Films with a high Metascore tend to also secure a high IMDb score.
    3. A Benchmark for the Viewer: The status of an "Oscar Winner" serves as a reliable quality mark, practically guaranteeing a high baseline level of satisfaction for the general audience.
5. Limitations of the Study
    - Inflation: Box office figures are presented in nominal dollars without adjusting for historical inflation. This distorts direct financial comparisons between modern films and those from previous decades.
    - Sample Bias: Dropping rows with missing values skewed the analysis toward modern cinema, as financial and rating data for older films is frequently unavailable in the OMDb database.
    - Statistical Outliers: Due to the strong right-skewness in revenue distribution (the presence of a few extreme box office hits), using the linear Pearson correlation may be insufficient. Future iterations of this analysis should apply logarithmic transformations to revenue data or utilize Spearman's rank correlation.